

Econometrics I

Lecture 4: Applied Regression Workshop: The Wage Equation

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Fall 2026

- Recall the Mincerian regression (wage equation):

$$\ln wage_i = \beta_0 + \beta_{ed}Education_i + \beta_{exp}Experience_i + \beta_{Fem}Female_i + \cdots + \varepsilon_i$$

- Let's revisit estimating this with the Cornwell and Rupert (PSID) data.

Process the data

```
suppressMessages(library(tidyverse))
suppressMessages(library(fixest))
suppressMessages(library(marginaleffects))

# first, the Cornwell and Rupert regression
data <- read.csv('./cornwell-rupert.csv') %>%
  mutate(ED_LEVEL=cut(ED,c(0,8,11,12,15,16,17),
    labels = c("NOHS", "SOMEHS", "HS", "SOMECOL", "COL", "POST"),
    right=TRUE))

# check that we did it correctly
table(data$ED,data2$ED_LEVEL)
```

```
reg_1 <- feols(LWAGE ~ ED + EXP + I(EXP^2) + WKS + OCC + SOUTH + SMSA
+ MS + UNION + FEM, data = data)

# dropping the constant
reg_2 <- feols(LWAGE ~ -1 + i(ED_LEVEL) + EXP + I(EXP^2) + WKS + OCC +
SOUTH + SMSA + MS + UNION + FEM, data = data2)

# not dropping the constant -- which category is omitted?
reg_3 <- feols(LWAGE ~ 1 + i(ED_LEVEL) + EXP + I(EXP^2) + WKS + OCC +
SOUTH + SMSA + MS + UNION + FEM, data = data2)

# change the omitted category -- how do coefficients change?
reg_4 <- feols(LWAGE ~ 1 + i(ED_LEVEL, ref="COL") + EXP + I(EXP^2) +
WKS + OCC + SOUTH + SMSA + MS + UNION + FEM, data = data2)

etable(list(reg_1, reg_2, reg_3, reg_4), export='./table_1.png')
```

Notice I've used the $i(\cdot)$ command to make categorical variables into dummies and $I(\cdot)$ to do polynomial terms.

Dependent Variable:	LWAGE			
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	5.245*** (0.0717)		5.655*** (0.0634)	6.161*** (0.0597)
ED	0.0565*** (0.0026)			
EXP	0.0404*** (0.0022)	0.0410*** (0.0022)	0.0410*** (0.0022)	0.0410*** (0.0022)
I(I(EXP ²))	-0.0007*** (4.78 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)
WKS	0.0045*** (0.0011)	0.0046*** (0.0011)	0.0046*** (0.0011)	0.0046*** (0.0011)
OCC	-0.1405*** (0.0147)	-0.1386*** (0.0151)	-0.1386*** (0.0151)	-0.1386*** (0.0151)
SOUTH	-0.0721*** (0.0125)	-0.0762*** (0.0126)	-0.0762*** (0.0126)	-0.0762*** (0.0126)
SMSA	0.1390*** (0.0121)	0.1436*** (0.0121)	0.1436*** (0.0121)	0.1436*** (0.0121)
MS	0.0674*** (0.0206)	0.0692*** (0.0207)	0.0692*** (0.0207)	0.0692*** (0.0207)
UNION	0.0901*** (0.0129)	0.0940*** (0.0130)	0.0940*** (0.0130)	0.0940*** (0.0130)
FEM	-0.3892*** (0.0252)	-0.3819*** (0.0253)	-0.3819*** (0.0253)	-0.3819*** (0.0253)
ED_LEVEL = NOHS		5.655*** (0.0634)		-0.5066*** (0.0284)
ED_LEVEL = SOMEHS		5.795*** (0.0624)	0.1400*** (0.0249)	-0.3666*** (0.0236)
ED_LEVEL = HS		5.903*** (0.0609)	0.2482*** (0.0229)	-0.2584*** (0.0194)
ED_LEVEL = SOMECOL		5.991*** (0.0610)	0.3364*** (0.0268)	-0.1702*** (0.0206)
ED_LEVEL = COL		6.161*** (0.0597)	0.5066*** (0.0284)	
ED_LEVEL = POST		6.188*** (0.0589)	0.5337*** (0.0295)	0.0271 (0.0213)
<i>Fit statistics</i>				
Observations	4,165	4,165	4,165	4,165
R ²	0.41826	0.41724	0.41738	0.41738
Adjusted R ²	0.41686	0.41527	0.41541	0.41541

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# dropping the constant
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etable(list(reg_1, reg_2, reg_3, reg_4), export='./table_1.png')
```

Note on interpreting effects with $\log(y_i)$: a dummy multiplies y by $\exp(\beta)$, and $\exp(\beta) \approx 1 + \beta$ only for small β :

► $\exp(-.3892) = .6776$

► $\exp(.05654) = 1.058$

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etable(list(reg_1,reg_2,reg_3,reg_4), export='./table_1.png')
```

- In Regression #1, what does the coefficient β_{ED} mean?
- What about in Regression #2? What is the interpretation of $\beta_{somecol}$?
- How about in Regression #3 and #4?

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Caveat: These Are Panel Data

- ▶ Cornwell–Rupert is 595 workers observed for 7 years, not 4,165 independent draws. A worker's errors are correlated across years (ability, occupation, location), and ED and FEM never change within a worker: the worst case of the Moulton factor from Lecture 3a ($\rho_x = 1$).
- ▶ Heteroskedasticity-robust SEs do not fix this. Clustering by worker roughly doubles them:

	ED	FEM	UNION	EXP
Coefficient	0.0565	−0.3892	0.0901	0.0404
SE: IID	0.0026	0.0252	0.0129	0.0022
SE: HC1	0.0027	0.0240	0.0125	0.0022
SE: clustered by worker	0.0056	0.0482	0.0240	0.0042

- ▶ In R: `cluster = ~ID` with `ID = rep(1:595, each = 7)` (rows are sorted by worker). Conclusions survive, but today's SEs are too small; panel data get their own treatment in Lecture 8a.

Formulating Linear Hypotheses

```
> print(linearHypothesis(reg_5, c("FEM = MALE"),  
+                          vcov = vcovHC(reg_5, type = "HC1")))
```

Linear hypothesis test:

FEM - MALE = 0

Model 1: restricted model

Model 2: LWAGE ~ -1 + ED + EXP + I(I(EXP^2)) + WKS + OCC + SOUTH + SMSA +
MS + UNION + FEM + MALE

Note: Coefficient covariance matrix supplied.

	Res.Df	Df	Chisq	Pr(>Chisq)
1	4155			
2	4154	1	263.33	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
> print(linearHypothesis(reg_6, c("FEM = 0"),  
+                          vcov = vcovHC(reg_6, type = "HC1")))
```

Linear hypothesis test:

FEM = 0

Model 1: restricted model

Model 2: LWAGE ~ 1 + ED + EXP + EXP2 + WKS + OCC + SOUTH + SMSA + MS +
UNION + FEM

Note: Coefficient covariance matrix supplied.

	Res.Df	Df	Chisq	Pr(>Chisq)
1	4155			
2	4154	1	263.33	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We can test the same hypothesis with an F -test two different ways:

- ▶ $H_0 : \beta_M = \beta_F$ if we omit the constant.
- ▶ $H_0 : \beta_F = 0$ if we include the constant.

Correlation in F-Tests

```
# Number of observations
n <- 200
# Generate random data
# Random integer values between 18 and 65
AGE <- sample(22:65, n, replace = TRUE)
# same as age-22 but one less for some observations
EXP <- AGE - 22 - rbinom(n=n,size=1,prob=0.4)
EXP[EXP < 0] <- 0 # replace negative values with 0
LWAGE = 2.5 + .02*AGE + .03*EXP + .5*rnorm(n)
# create data frame
df <- data.frame(LWAGE,AGE,EXP)

# estimate OLS
reg <- feols(LWAGE ~ AGE + EXP, data = df)
summary(reg)
linearHypothesis(reg,c("AGE+EXP=0"),
  vcov = vcovHC(reg, type = "HC1"))
```

```
> reg <- feols(LWAGE ~ AGE + EXP, data = df) # estimate OLS
> summary(reg)
OLS estimation, Dep. Var.: LWAGE
Observations: 200
Standard-errors: IID

```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.082345	1.630139	1.277404	0.20296
AGE	0.040453	0.072772	0.555890	0.57892
EXP	0.005372	0.072809	0.073781	0.94126

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
RMSE: 0.495771 Adj. R2: 0.585937
> linearHypothesis(reg,c("AGE+EXP=0"),
+                   vcov = vcovHC(reg, type = "HC1"))

Linear hypothesis test:
AGE + EXP = 0

Model 1: restricted model
Model 2: LWAGE ~ AGE + EXP

Note: Coefficient covariance matrix supplied.
```

	Res.Df	Df	Chisq	Pr(>Chisq)
1	198			
2	197	1	317.52	< 2.2e-16 ***

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Notice that t -stats are not significant but F -test is huge, why?

Marginal Effects

Our regression specification contained both linear terms and quadratic terms for experience

$$\ln wage_i = \beta_0 + \beta_{exp} Experience_i + \beta_{exp^2} Experience_i^2 + X_i' \beta + \varepsilon_i$$

We can compute the marginal effect of an additional year of experience as:

$$\frac{\partial \ln wage_i}{\partial Experience_i} = \beta_{exp} + 2\beta_{exp^2} Experience_i \equiv g(\beta).$$

- ▶ Note: We cannot simply interpret β_{exp} or β_{exp^2} on their own.
- ▶ To compute standard errors on marginal effects, we can't just look at the t -stats.
- ▶ This is also an issue in **nonlinear models** like Probit and Logit (later).

- Suppose we have an asymptotic distribution for an estimator:

$$\sqrt{n}(\mathbf{b} - \beta) \Rightarrow_d \mathcal{N}(\mathbf{0}, \Sigma).$$

- Then the asymptotic distribution of a function of the estimator is

$$\sqrt{n}(g(\mathbf{b}) - g(\beta)) \Rightarrow_d \mathcal{N}\left(\mathbf{0}, (\nabla g(\beta))' \Sigma \nabla g(\beta)\right),$$

where $\nabla g(\beta)$ is the gradient of $g(\beta)$:

$$\nabla g(\beta) = \left(\frac{\partial g(\beta)}{\partial \beta_1}, \frac{\partial g(\beta)}{\partial \beta_2}, \dots, \frac{\partial g(\beta)}{\partial \beta_K} \right)^T.$$

- Note that we can estimate $\nabla g(\beta)$ with $\nabla g(\mathbf{b})$.
- Notice how the covariance between the coefficients Σ matters!
- What is the expression for the standard error of the marginal effect of experience?

Delta Method / Marginal Effects in R

```
>
> library(marginaleffects)
> avg_slopes(reg_2, variables = "EXP", vcov = "HC1")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.0134	0.000591	22.8	<0.001	378.7	0.0123	0.0146

Term: EXP
Type: response
Comparison: dY/dX

Here we get the answer **correct**

What is the exercise we perform in each case?

```
> # don't use the I() to construct interactions
> reg_wrong <- feols(LWAGE ~ -1 + i(ED_LEVEL) + EXP + EXP2 + WKS + OCC +
+                      SOUTH + SMSA + MS + UNION + FEM,
+                      data = data %>% mutate(EXP2 = EXP^2))
> avg_slopes(reg_wrong, variables = "EXP", vcov = "HC1")
```

Estimate	Std. Error	z	Pr(> z)	S	2.5 %	97.5 %
0.041	0.0022	18.6	<0.001	254.8	0.0367	0.0453

Term: EXP
Type: response
Comparison: dY/dX

Here we get the answer **wrong**

Simulating the Asymptotic Distribution

- ▶ Given the asymptotic distribution of a parameter estimate

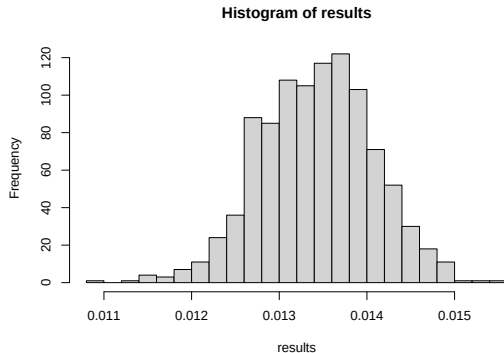
$$\mathbf{b} \stackrel{a}{\sim} \mathcal{N}(\boldsymbol{\beta}, \boldsymbol{\Sigma}/n),$$

we have an estimated density function $\hat{f}$. Let $\hat{f}$ be the multivariate normal density with mean $\mathbf{b}$ and variance $\hat{\boldsymbol{\Sigma}}/n$.

- ▶ We can simulate the asymptotic distribution of $g(\mathbf{b})$ by
 - Simulating draws $\mathbf{b}_m$ for $m = 1, 2, \dots, M$ from $\hat{f}$
 - Computing $g(\mathbf{b}_m)$ for each draw
 - Then $(g(\mathbf{b}_1), g(\mathbf{b}_2), \dots, g(\mathbf{b}_M))$ will be a simulated asymptotic distribution for $g(\mathbf{b})$
- ▶ This can be useful when you have code to compute $g(\cdot)$, but computing the derivative of $g(\cdot)$ would be difficult. For example, when $g(\cdot)$ represents a complex behavioral (or equilibrium) model.
- ▶ Also if you are too lazy to load marginal effects

Bootstrap Marginal Effects in R

```
# quick bootstrap comparison
n <- dim(data)[1]
results <- rep(0,1000)
for (i in 1:1000){
  my_weights <- rmultinom(1,n,rep(1/n,n))/n
  my_reg <- feols(LWAGE ~ -1 + i(ED_LEVEL) + EXP + I(EXP^2) + WKS + OCC +
    SOUTH + SMSA + MS + UNION + FEM,
    data = data,
    # multiplier bootstrap
    weights=my_weights)
  results[i] <- my_reg$coefficients[7] +
    sum(2 * my_reg$coefficients[8] * my_weights * data$EXP)
}
print(mean(results))
# [1] 0.01342801
print(quantile(results,c(0.025,.975)))
#2.5%      97.5%
# 0.01219617 0.01463365
hist(results,20)
```



How does the distribution of results compare to delta method? Can we do better?

Heterogeneous Effects

- ▶ When we have a model of the form

$$Y_i = \beta_0 + \beta_1 X_{1i} + \varepsilon_i$$

we're implicitly saying that the effect of X_1 is the same for all individuals.

- ▶ Often we would like to relax this, allowing different groups to have different slopes with respect to X_1 .
- ▶ This is easy as long as the group membership is observed in the data. We simply interact the regressor with dummy variables:

$$Y_i = \beta_0 + \beta_{0F} D_{Fi} + \beta_1 X_{1i} + \beta_2 X_{1i} D_{Fi} + \varepsilon_i$$

where D_{Fi} is a dummy variable for whether individual i is female. Note that we have allowed for the intercepts and slopes to vary by sex here.

- ▶ Run this on your own and experiment with `i(·)` operator and `:` and `*`.

Mincerian Regression: Measurement Error

- ▶ What happens if one of the variables of interest is measured with error?
- ▶ Let's say the recorded education might be one year more or less than the person's actual education.
- ▶ Note: this may already be happening in the data, but let's make it happen more.

```
set.seed(93)
noise <- sample(-1:1,dim(data)[1],replace=T)

reg_8 <- feols(LWAGE ~ ED_NOISY + EXP + I(EXP^2) + WKS + OCC + SOUTH + SMSA
+ MS + FEM + UNION, data = data %>% mutate(ED_NOISY = ED + noise))
summary(reg_8)

etable(list(reg_1,reg_8), export='./table_noise.png')
```

Dependent Variable: LWAGE		
Model:	(1)	(2)
Variables		
Constant	5.245*** (0.0717)	5.377*** (0.0706)
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FEM	-0.3892*** (0.0252)	-0.3892*** (0.0254)
ED_NOISY		0.0487*** (0.0025)
Fit statistics		
Observations	4,165	4,165
R ²	0.41826	0.40837
Adjusted R ²	0.41686	0.40695
IID standard-errors in parentheses		
Signif. Codes: ***: 0.01, **: 0.05, *: 0.1		

Omitted Variables Bias, Revisited

- ▶ Suppose the econometrician only observes regressors $\mathbf{X}$, but the true model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}\gamma + \boldsymbol{\varepsilon},$$

- ▶ The OLS estimator will equal

$$\mathbf{b} = (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{y} = \boldsymbol{\beta} + (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\mathbf{z}\gamma + (\mathbf{X}'\mathbf{X})^{-1} \mathbf{X}'\boldsymbol{\varepsilon}$$

- ▶ The last term is mean zero given strict exogeneity, $\mathbb{E}[\boldsymbol{\varepsilon} \mid \mathbf{X}, \mathbf{z}] = \mathbf{0}$.
- ▶ Note that the second term will not be zero if $\gamma \neq 0$ and $\mathbf{z}$ is correlated with the (non-constant) columns of $\mathbf{X}$.
- ▶ Implication: correlation between omitted variables and the observed regressors makes OLS biased.

- Using the Frisch-Waugh theorem, we can show that

$$\mathbb{E} [b_{OLS,k} \mid \mathbf{X}, \mathbf{z}] = \beta_k + \gamma \left(\frac{\tilde{\mathbf{x}}_k' \mathbf{z}}{\tilde{\mathbf{x}}_k' \tilde{\mathbf{x}}_k} \right)$$

where $\tilde{\mathbf{x}}_k$ is the residual from regressing x_k on $\mathbf{X}_{-k}$, all the regressors besides x_k . The ratio is the slope from regressing $\mathbf{z}$ on $\tilde{\mathbf{x}}_k$: the partial covariance of z and x_k over the partial variance of x_k .

- Suppose positive correlation between regressor x_k and omitted variable z .
- Also suppose $\beta_k > 0$ and $\gamma > 0$ so both variables have positive effects.
- Let's compare the average value of the dependent variable for $x_k = 0$ and $x_k = 1$. Two things change between these points:
 - Dependent variable Y increases by β_k because of direct effect of x_k .
 - Value of z should be higher because of the positive correlation between x_k and z . Higher values of z also contribute to a higher dependent variable because $\gamma > 0$.

Omitted Variables Bias in Mincerian Regression

- What sort of variables might the wage equation omit, and how would you expect them to affect the estimated coefficients?

Dependent Variable:	LWAGE			
Model:	(1)	(2)	(3)	(4)
<i>Variables</i>				
Constant	5.245*** (0.0717)		5.655*** (0.0634)	6.161*** (0.0597)
ED	0.0565*** (0.0026)			
EXP	0.0404*** (0.0022)	0.0410*** (0.0022)	0.0410*** (0.0022)	0.0410*** (0.0022)
I(EXP ²)	-0.0007*** (4.78 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)	-0.0007*** (4.8 × 10 ⁻⁵)
WKS	0.0045*** (0.0011)	0.0046*** (0.0011)	0.0046*** (0.0011)	0.0046*** (0.0011)
OCC	-0.1405*** (0.0147)	-0.1386*** (0.0151)	-0.1386*** (0.0151)	-0.1386*** (0.0151)
SOUTH	-0.0721*** (0.0125)	-0.0762*** (0.0126)	-0.0762*** (0.0126)	-0.0762*** (0.0126)
SMSA	0.1390*** (0.0121)	0.1436*** (0.0121)	0.1436*** (0.0121)	0.1436*** (0.0121)
MS	0.0674*** (0.0206)	0.0692*** (0.0207)	0.0692*** (0.0207)	0.0692*** (0.0207)
UNION	0.0901*** (0.0129)	0.0940*** (0.0130)	0.0940*** (0.0130)	0.0940*** (0.0130)
FEM	-0.3892*** (0.0252)	-0.3819*** (0.0253)	-0.3819*** (0.0253)	-0.3819*** (0.0253)
ED_LEVEL = NOHS		5.655*** (0.0634)		-0.5066*** (0.0284)
ED_LEVEL = SOMEHS		5.795*** (0.0624)	0.1400*** (0.0249)	-0.3666*** (0.0236)
ED_LEVEL = HS		5.903*** (0.0609)	0.2482*** (0.0229)	-0.2584*** (0.0194)
ED_LEVEL = SOMECOL		5.991*** (0.0610)	0.3364*** (0.0268)	-0.1702*** (0.0206)
ED_LEVEL = COL		6.161*** (0.0597)	0.5066*** (0.0284)	
ED_LEVEL = POST		6.188*** (0.0589)	0.5337*** (0.0295)	0.0271 (0.0213)
<i>Fit statistics</i>				
Observations	4,165	4,165	4,165	4,165
R ²	0.41826	0.41724	0.41738	0.41738
Adjusted R ²	0.41686	0.41527	0.41541	0.41541

Let's look at the workbook now.

- ▶ Open `inclass_session4.Rmd` (in the Fall/L5 - Wage Equation Example folder of the course repo).
- ▶ Reproduce today's regressions, the delta method by hand, and the bootstrap.
- ▶ Then work the two problems with a neighbor:
 1. A union premium for whom?
 2. Attenuation, live